MTH-603 Final-Term BY Attiq Kundi

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(Updated at 09-09-2022)

1. Trapezoidal rule of integration of a define integral is of
   O(h2)
2. In multistep method, if we choose small step size
   Increase
3. tabular function but give more error
   Trapezoidal method
4. dy/dx=t+ $\sqrt{y}$ initial condition y_0=1 at t_0=0 using modified Euler's method, for the range 0≤t≤0.6 h=0.2
   1.2
5. points the maximum value of 2nd derivative f(x)=-(2/x)
   at x=4
6. dy/dx=t+y with initial condition y_0=1 at t_0=1 and h_0=0.1 them k_1 of RK
   o.2
7. accuracy of trapezoidal and Simpson's rule
   Richardson's extrapolation method
8. Adam mouton also known as
   Linear multistep
9. F(x) in the interval [a,b] gives
   Area under f(x) on [a,b]
10. dy/dx=t+ $\sqrt{y}$ initial condition y_0=1 at t_0=0 find 2nd term in Taylor series when t=1, y'=0.2 and h=0.1
    0.2

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11. while employing trapezoidal and Simpson rules to evaluate the double integral numerically, by using trapezoidal and Simpson rule

    plain region

12. $dy/dt=1-\sqrt{y}$ initial condition $y_0=1$ at $t_0=0$ find 2nd term in Taylor series when t=0.5, y'=0

    0

13. By the lines x=a, x=b, y=c, y=d we shall employ either trapezoidal or Simpson's rule, variable at time

    One

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14. dy/dt=t+ y with the initial condition y_0=1 at t_0=1 find the 2nd term in Taylor series when t=1.5, y'=1.2

    0.6

15. dy/dt=y-t/y=t with the initial condition y=1 at t=0 find the 3rd term in Taylor series when t=0, y''=0.2

    0.018

16. lead towards the numerical integration method

    All above choice are true

17. Adams-Bashforth is a multistep method

    True

18. To evaluate a define integral of tabular function f(x), linear approximate led to

    Trapezoidal method

19. points the minimum value of 2nd derivative f(x)=-(2/x)

    at x=1

20. in Trapezoidal rule, subinterval o equal length using—points

    n+1

21. in solving the differentiate equation by Euler's method y(0.2) is calculate as

    1.02

22. Adams method are superior if output at many point

    True

23. K4 is

    K4=hf(tn+h1yn+k3)

24. Yn+1=y_n+1/6(k_1+2 k_2+2k_3 +k_4) is

    Runge-Kutta's formula

25. To evaluate a define integral of tabular function f(x), piecewise quadratic approximate led to

    Simpson's rule

26. F(x) in continuous on [a,b] and we divide [a,b]

    even

27. dy/dx=y+x /y-x with initial condition y_0=1 at x_0=0 and h=0.2 them k_2 of Runge-Kutta method

    o.22

28. dy/dx=t+y with initial condition y_0=1 at t_0=1 find the 3rd Taylor series when t=1.5 and y'=1 and y''=1.2

    o.1

29. K3 is

    K3=hf(tn+h/2,yn+k2/2)

30. trapezoidal and Simpson's integration are just linera combination values

    independent

31. the area of trapezoid is obtained by adding the area of

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Rectangle

32. dy/dx=$xy^{1/2}$ with initial condition y_0=1 at x_0=1 and h_0=0.1 them k_1 of Runge-Kutta

    o.1

33. dy/dt=y-t/y+t initial condition y_0=1 at t_0=0 find 2nd term in Taylor series when t=0.2, y'=0.3

    0.06

34. dy/dx=t+ y with initial condition y0=1 at t0=1 find the 3 rd Taylor series when t=1 and y'=0.2 and y''=2 and h=0.1

    1

35. multiset method is that the local truncation error(L.T.E)

    True

36. simpson's rule is a numerical method that approximate polynomial

    Quadric

37. in simpson's rule, we can estimate the integral by----

    adding

38. trapezoidal and Simpson rule with respect to ----

    single

39. dy/dt=t+y with initial condition y0=1 at t0=0 find the 3rd Taylor series when t=0.5 and y''=0.25

    0.01325

40. dy/dt=1- y initial condition y0=1 at t0=0 find 2nd term in Taylor series when t=0.5, y'=0

    0

41. Yn+1=yn+1/6(k1+2 k2+2k3 +k4) k1 is

    K1=hf(tn,yn)

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